Swiss Institute of Bioinformatics

PR STATS

Metagenomic Data Analysis of Microbial Communities (MGMC01)

Day 1 - Session 1

PhD Jeferyd Yepes-García
University of Fribourg
October 26th, 2026

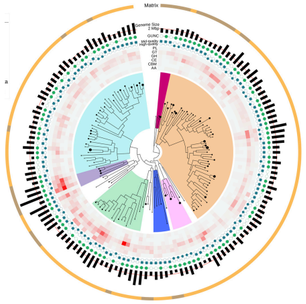
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About Me



nextflow
AMBASSADOR



TaxoFlow
The tutorial

nf-core/mag

MAGFlow
Workflow for MAG quality assessment and taxonomical annotation

2P

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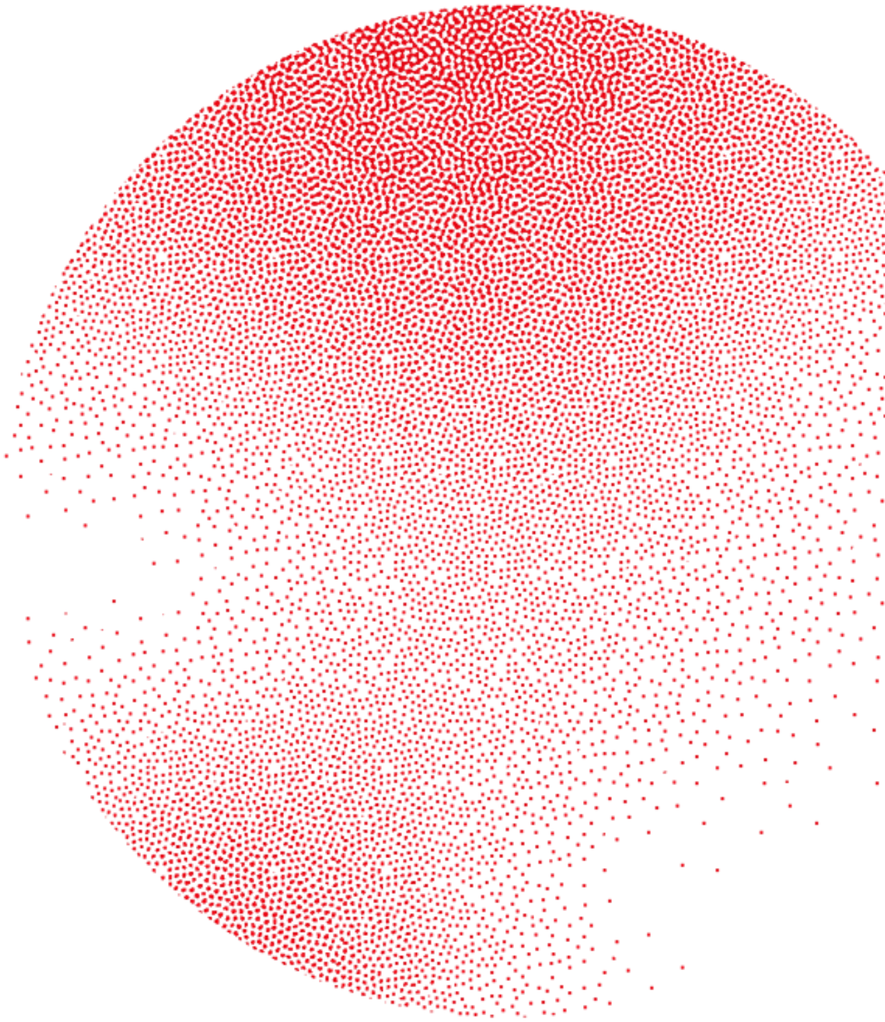
Course schedule

26–30 October 2026 · 14:00–19:30 UK time (GMT) · Live sessions recorded

Day 1 · Mon 26 Oct	Day 2 · Tue 27 Oct	Day 3 · Wed 28 Oct	Day 4 · Thu 29 Oct	Day 5 · Fri 30 Oct
14:00–17:30 | **Introduction to metagenomics**	14:00–17:30 | **Targeted sequencing: theory**	14:00–17:30 | **Whole-genome sequencing and taxonomic classification**	14:00–17:30 | **Metagenome assembly**	14:00–17:30 | **MAG quality, annotation and classification**
Break 17:30–18:00	Break 17:30–18:00	Break 17:30–18:00	Break 17:30–18:00	Break 17:30–18:00
18:00–19:30 | **Quality control and contamination assessment**	18:00–19:30 | **Targeted sequencing: downstream analyses**	18:00–19:30 | **Taxonomic profiling and downstream analysis**	18:00–19:30 | **Binning and MAGs**	18:00–19:30 | **FAIR data and reproducibility**

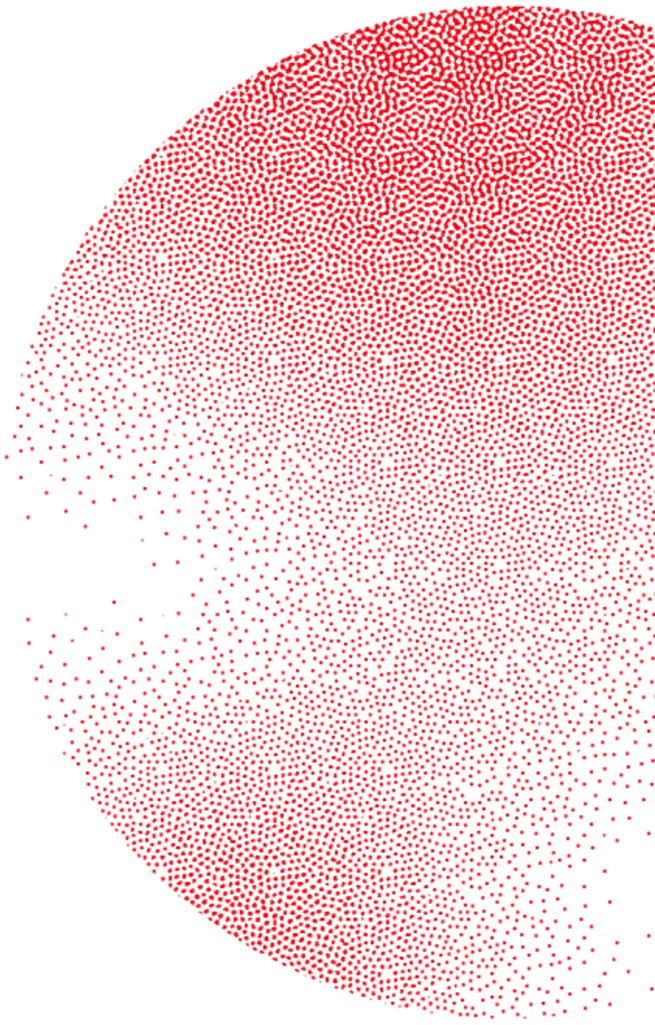
This presentation

- Learning outcomes
- Context: Microbiome and metagenomics
- Approaches to study the metagenome
- Shotgun sequencing
- Targeted sequencing
- Short reads vs long reads
- Resources
- Summary



Learning outcomes

- Explain the core concepts, terminology and scope of metagenomics.
- Describe the biological and ecological rationale for studying microbial communities *in situ*, including the concept of the “unculturable majority”.
- Discuss key considerations in experimental design for metagenomic studies, including replication, controls and confounding variables.

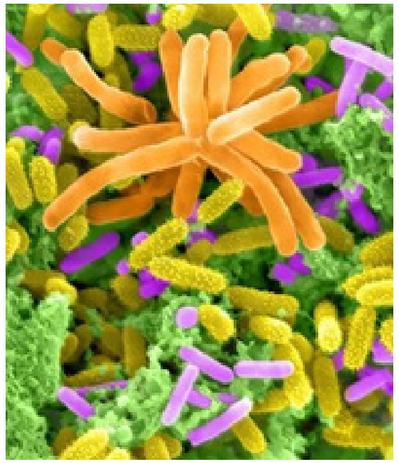


What is metagenomics?

“The study of the DNA of uncultured organisms”

> 99% of all microbes do not grow *in vitro*

Think twice!



Methylamine-enriched community of Lake Washington sediment featuring Methyloteneracells Photo © Dennis Kunkel Microscopy, Inc. (Color by Ekaterina Latypova)

Some definitions…

A genome:

Entire genetic information of a single organism

A metagenome:

Entire genetic information of an assemblage/mixture of organisms

A metatranscriptome:

Entire collection of transcripts of an assemblage/mixture at a given time

What is a microbiome?

The microbiome (or microbiota):
is an ecological community of commensal, symbiotic and pathogenic microorganisms found in and on all multicellular organisms studied to date from plants to animals. A microbiota includes bacteria, archaea, protists, fungi and viruses.



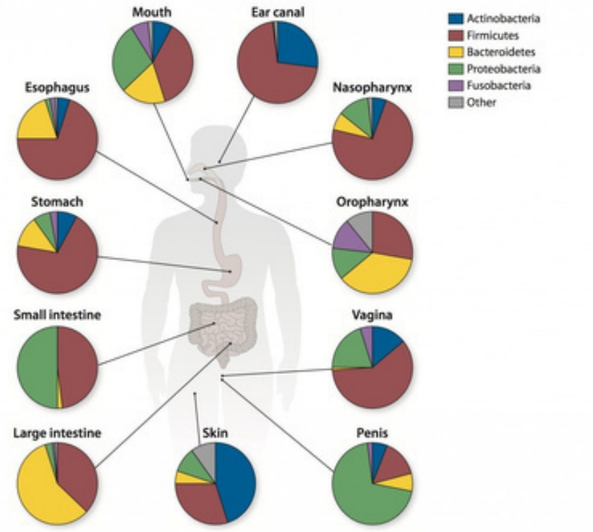


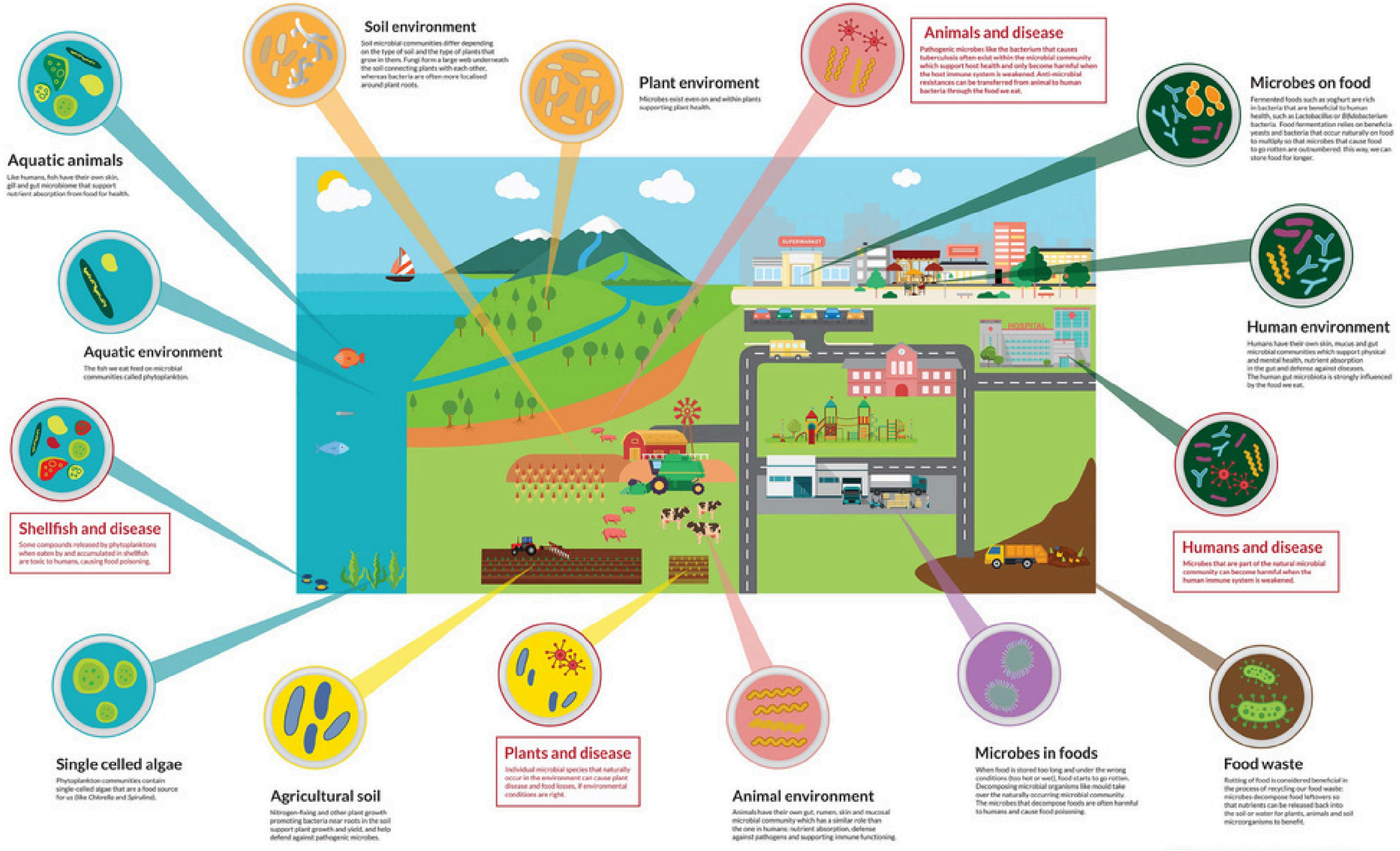
Figure 1
General patterns in the composition of bacterial communities in various body habitats. Pie charts illustrate percentage of 16S rRNA gene sequences representing the dominant bacterial phyla. Data compiled from Charlson et al. 2010, Costello et al. 2009, Dicksved et al. 2009, Frank et al. 2003, Hayashi et al. 2005, Kim et al. 2009, Pei et al. 2004, Price et al. 2010.

Flora or microflora synonym should not be used in scientific literature. It can be found in popular literature or in a yogurt/probiotic advertisement destined to the general public.

Microbes are everywhere in the food system

Diverse microbial communities consisting of fungi, bacteria, protozoa and other micro-organisms occur in all parts of our food system and are essential in its functioning and health, for food security and climate change mitigation.

Individual microbes can be harmful to plant, animal and human health if environmental conditions are in their favour. These microbes are often a natural part of microbial communities in low numbers.



Microbiome

IN NUMBERS

100 Trillion
symbiotic microbes live in and on every person and make up the human microbiota

The human body has more microbes than there are stars in the milky way

95%
of our microbiota is located in the GI tract

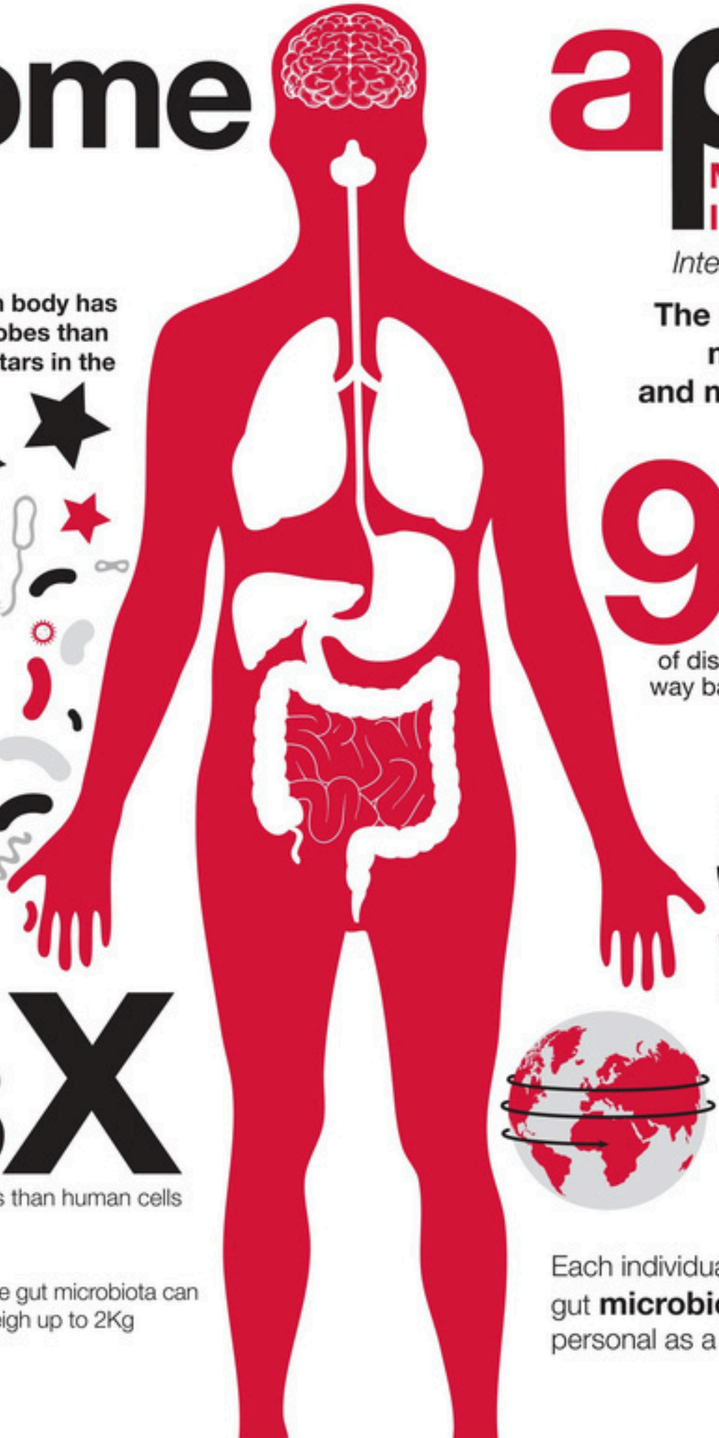
150:1
The genes in your microbiome outnumber the genes in our genome by about 150 to one

The surface area of the **GI tract** is the same size as 2 tennis courts

You have 1.3X
more microbes than human cells

>10,000
Number of different microbial species that researchers have identified living in and on the human body

2kg
The gut microbiota can weigh up to 2Kg



apc Microbiome Ireland
Interfacing Food & Medicine

The microbiome is more medically accessible and manipulable than the human genome

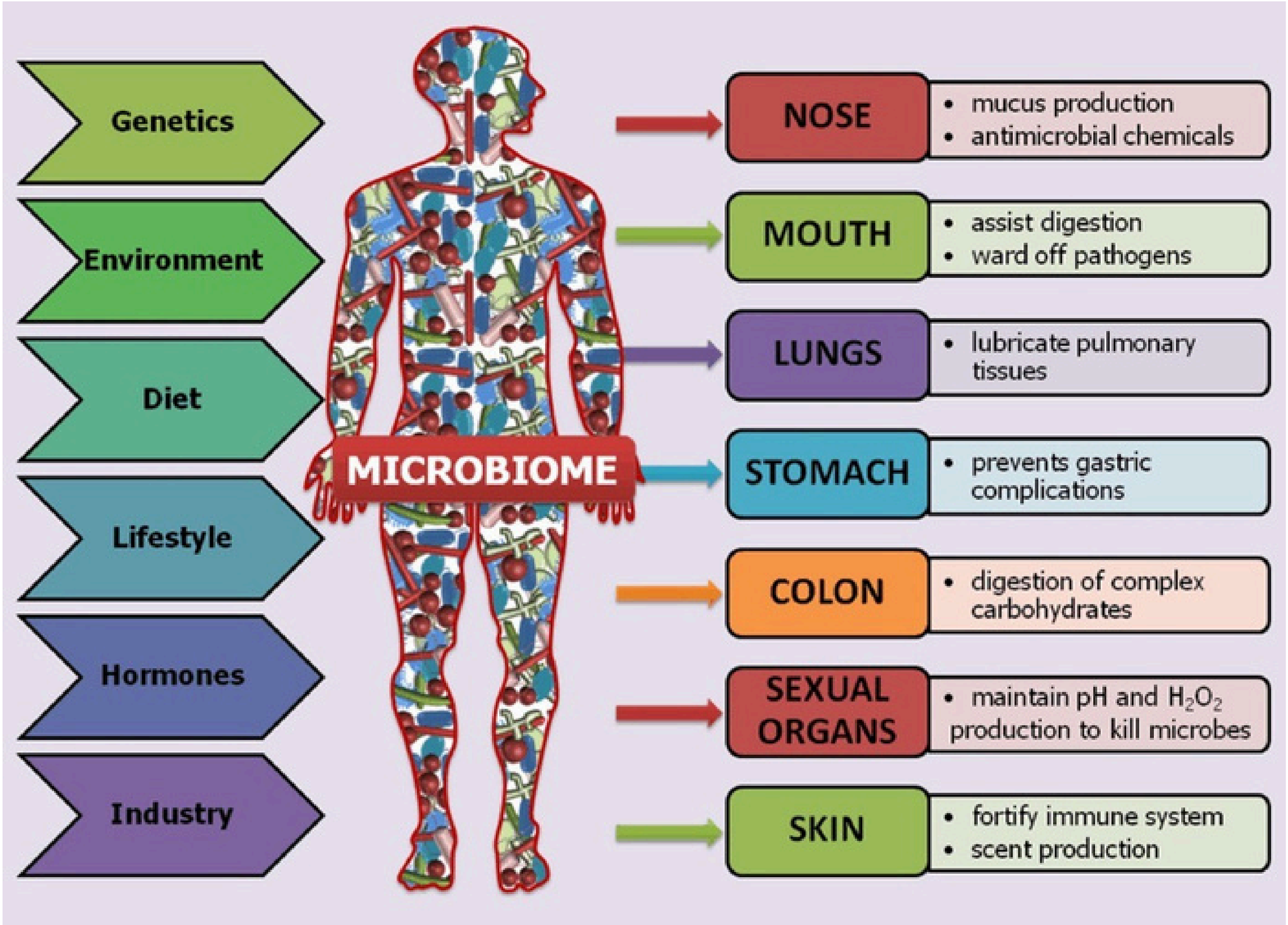
It is thought that **90%** of disease can be linked in some way back to the gut and health of the microbiome

5:1
Viruses:Bacteria
in the gut microbiota

2.5 The number of times your body's microbes would circle the earth if positioned end to end

Each individual has a unique gut **microbiota**, as personal as a fingerprint




Genetics
Environment
Diet
Lifestyle
Hormones
Industry
MICROBIOME
NOSE
• mucus production
• antimicrobial chemicals
MOUTH
• assist digestion
• ward off pathogens
LUNGS
• lubricate pulmonary tissues
STOMACH
• prevents gastric complications
COLON
• digestion of complex carbohydrates
SEXUAL ORGANS
• maintain pH and H_2O_2 production to kill microbes
SKIN
• fortify immune system
• scent production

Examples of roles and impact of the microbiome in the human gut

Benefits

- Digestion, production of vitamins & antioxidants, iron absorption, fighting pathogens

Diseases

- Obesity, diabetes, some cancers (*Faecalibacterium prausnitzii*), heart disease

Immunity

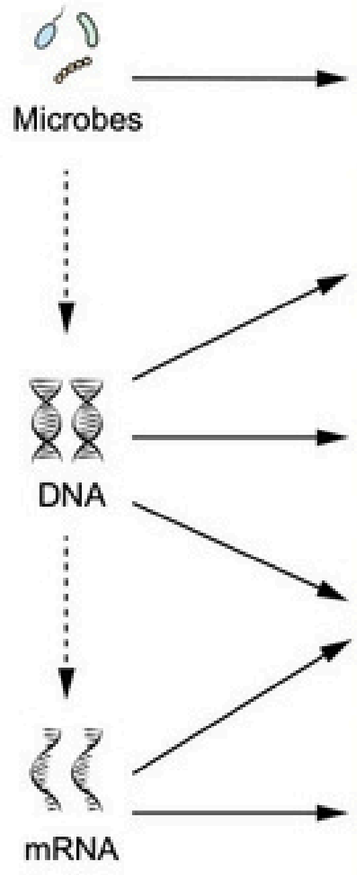
- Eczema & allergy, asthma, autoimmune disorders

Behavior

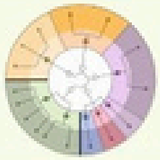
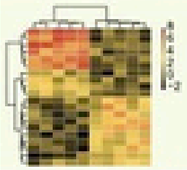
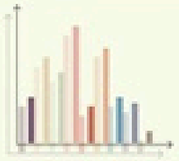
- Psychiatric disorders (depression), autism

How to study a microbiome/metagenome?

A



B

Method	Advantages	Limitations
 Culturome	• High-throughput • Targeted selection • Provides microbial isolates	• Expensive • Laborious • Influenced by media and the environment
 Amplicon (16S/18S/ITS)	• Quick analysis • Low-biomass requirement • Applicable to samples contaminated by host DNA	• PCR and primer biases • Resolution limited to genus level • False positive in low-biomass samples
 Metagenome	• Taxonomic resolution to species or strain level • Functional potential • Uncultured microbial genome	• Expensive • Time-consuming in analysis • Host-derived contamination
 Virome	• Can identify RNA and DNA viruses • Quick diagnosis	• Most expensive • Difficult to analysis • Severe host-derived contamination
 Metatranscriptome	• Can identify live microbes • Can evaluate microbial activity • Transcript-level responses	• Complex sample collection and analysis • Expensive and complex in sequencing • Host mRNA and rRNA contamination

Liu et al. (2021)

The choice of the method depends on 3 main questions

Who is there? Taxonomic profiling

catalogue of species, diversity,genus, etc.

distribution (how many of each)

What are they doing? Functional analysis

genes/proteins, GO terms, metabolic pathways

functional annotation, bioprospection

How do they compare? Differential analysis

pairwise or multiple comparisons

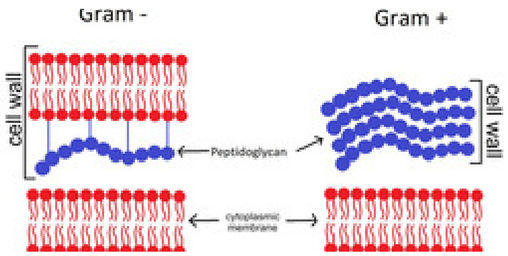
correlation with environmental factors

In addition: Meta-analysis

combining multiple studies



- easy + hard



What is challenging?

Lab challenges:

Different DNA extraction steps can give different sequence sets

Data analysis challenges:

Huge amount of sequences to process

Most sequences are absent from the existing reference databases

Assembly of communities can be hard with short reads

No perfect tool or workflow for analysis exists

Different analysis tools can give different results

Even worse, the same version of a tool can give different results depending on the reference database used

Technical options for metagenome/microbiome sequencing

Long reads vs short reads?

Read type	Advantage	Disadvantage
Short (Illumina)	Cheaper to produce More sensitive	Many biases
Long (Nanopore)	Longer sequences	More expensive High error rate
Long (PacBio)	Longer sequences HiFi reads	Less counts

Targeted vs Whole Genome Shotgun (WGS)

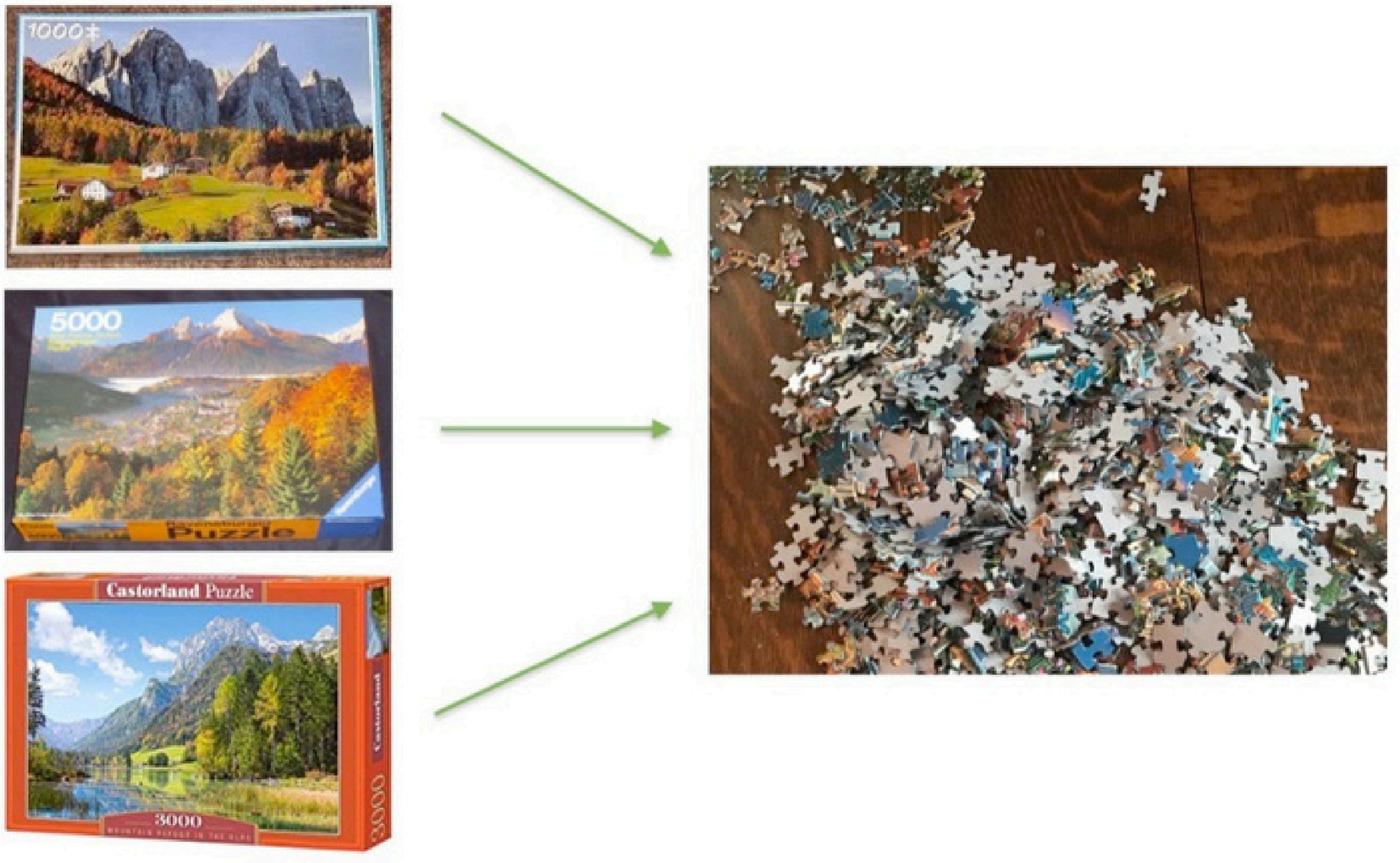
Amplicon (targeted)	WGS
Sequence alignment to a reference database is sufficient	Requires *de novo* assembly
PCR selection of a gene of interest	No selection
16S or ITS or COI or other…	all DNA is sequenced (contaminants!)
Error correction (denoising)	QC/cleaning + *de novo* assembly
Clustering to ASV or OTU	Binning to MAG
Taxonomic assignment	Taxonomic assignment
No function available	Functional assignment
Downstream analysis	Downstream analysis
E.g., alpha and beta diversity	E.g., comparative genomics

Metagenomics via Whole Genome Shotgun (WGS)

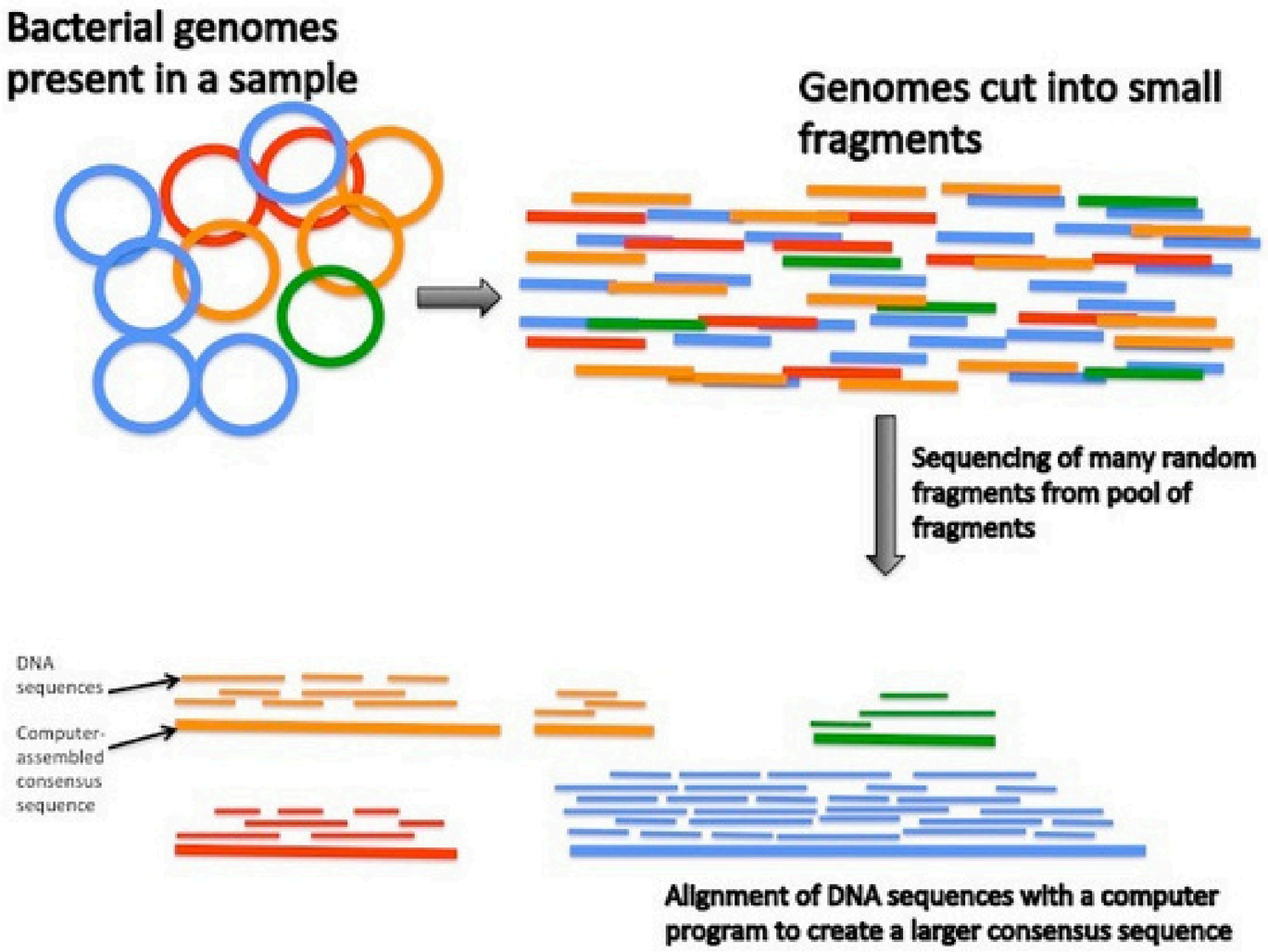
Difficulties:

- Requires *de novo* assembly
- Many genomes (hundreds to thousands)
- Often closely related
- Coverage very different (abundant vs rare)
- Large amounts of data
- Host contamination

Like thousands jigsaw puzzles mixed together!!



Whole metagenome assembly



After genome assembly

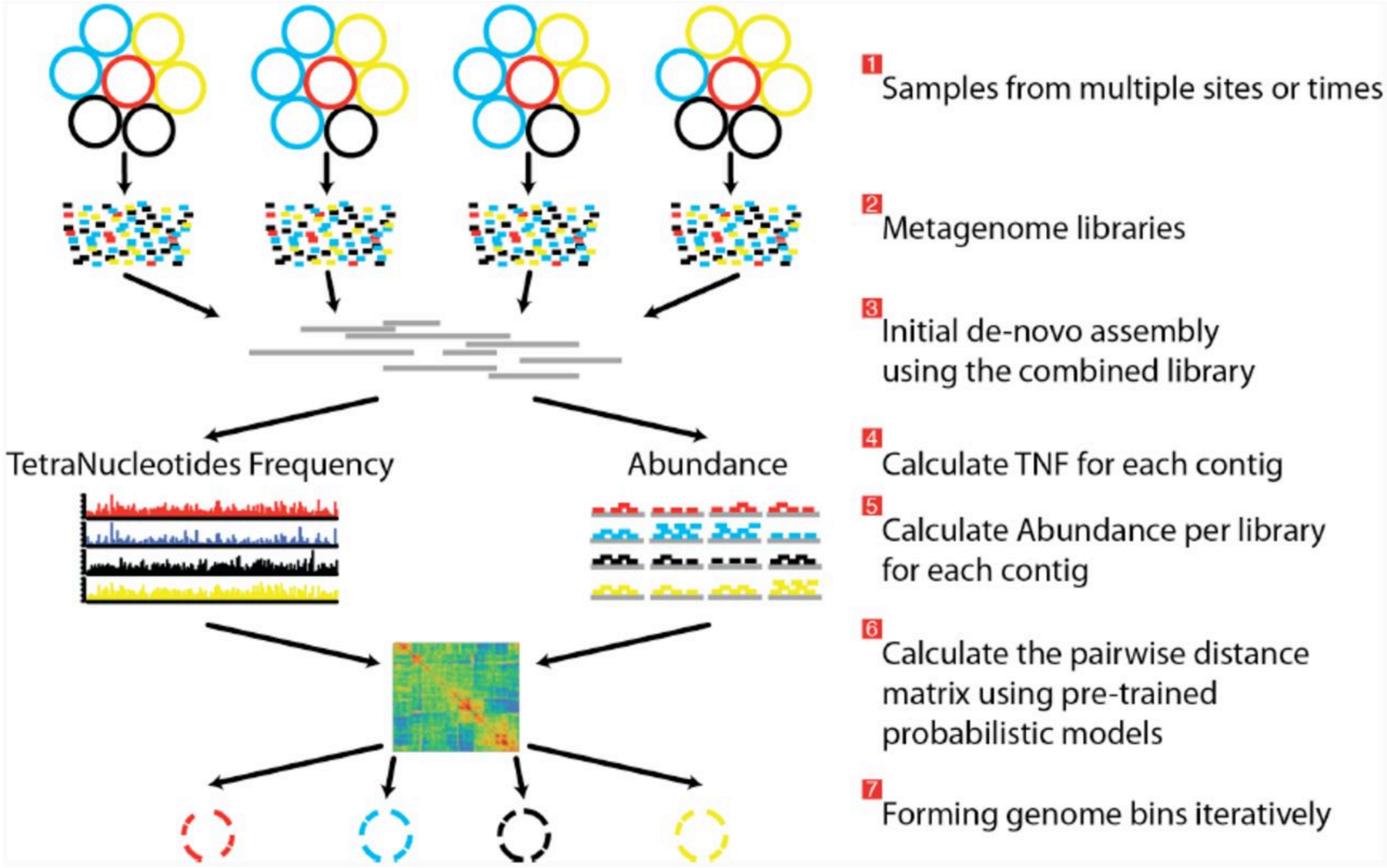
- Most metagenomics assemblers use de Bruijn graphs
- Even single genome assemblies are often still highly fragmented
- Metagenome assemblies are worse as many methods used for reducing the fragmentation cannot be applied in a mixture of genomes

- The idea is to sort the data into smaller groups or «bins»



Assembly and binning

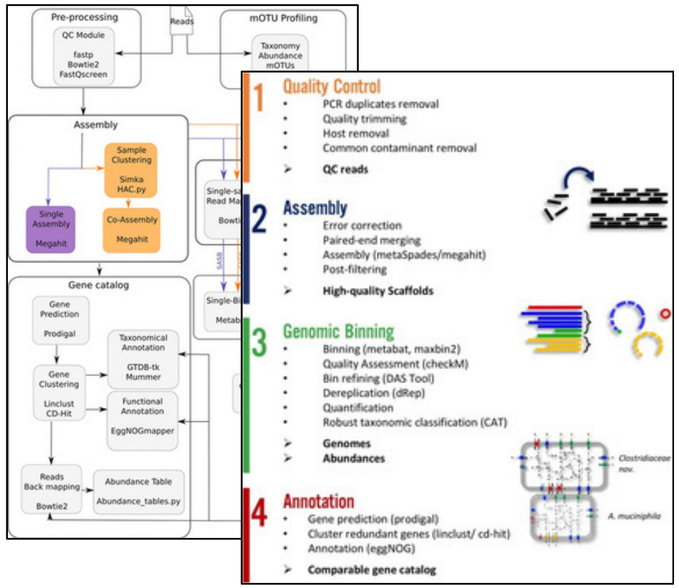
Binning can use various data to produce Metagenome Assembled Genomes (MAGs)



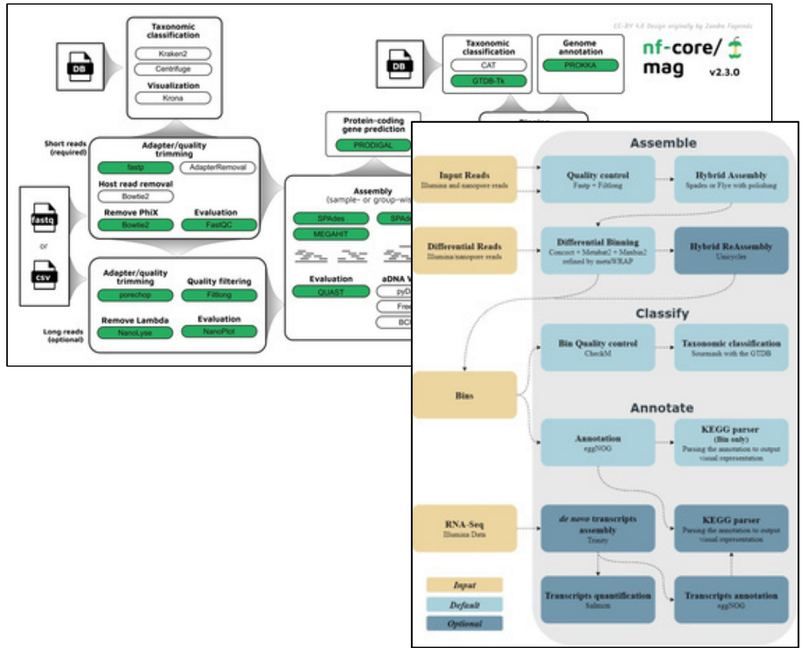
Complete automated workflows and pipelines

Most use Snakemake or Nextflow or custom pipelines

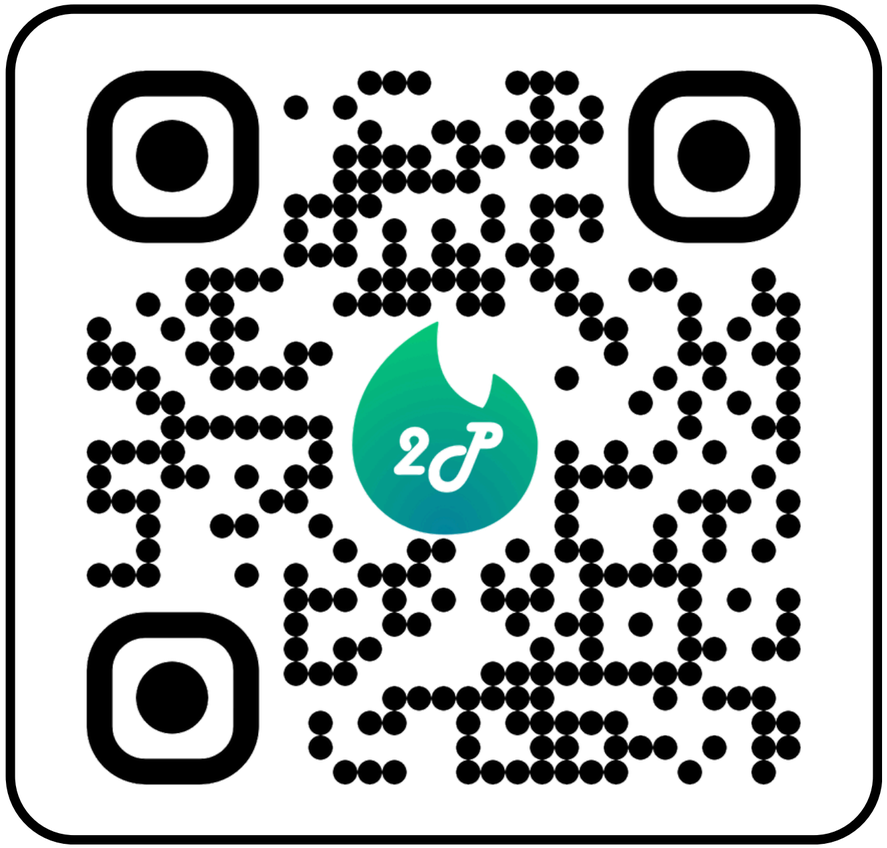
snakemake



nextflow



Automated workflows and pipelines



2pipe.app/

Targeted vs Whole Genome Shotgun (WGS)

Amplicon (targeted)	WGS
Sequence alignment to a reference database is sufficient	Requires *de novo* assembly
PCR selection of a gene of interest	No selection
16S or ITS or COI or other…	all DNA is sequenced (contaminants!)
Error correction (denoising)	QC/cleaning + *de novo* assembly
Clustering to ASV or OTU	Binning to MAG
Taxonomic assignment	Taxonomic assignment
No function available	Functional assignment
Downstream analysis	Downstream analysis
E.g., alpha and beta diversity	E.g., comparative genomics

Targeted or Amplicon sequencing

Library preparation requires PCR amplification

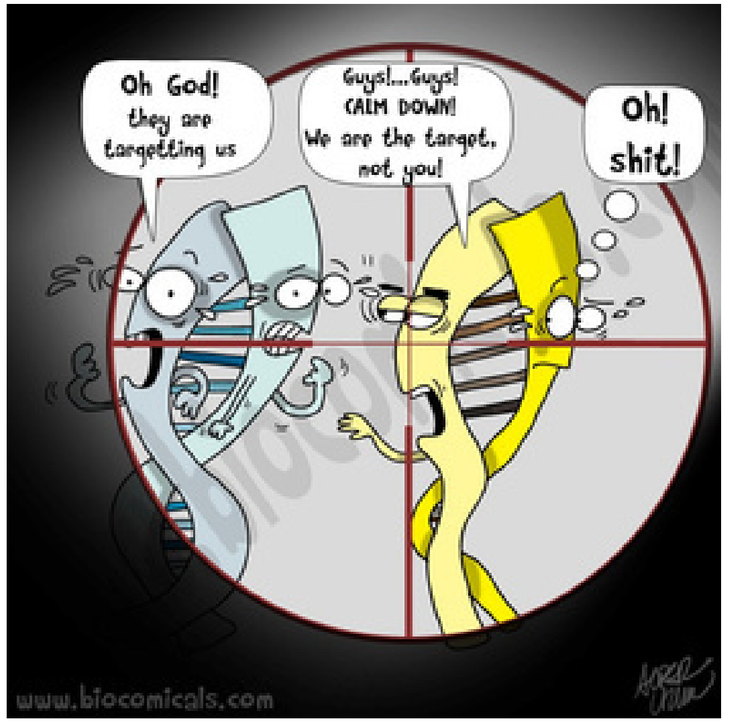
QC (Clean reads)

optional: combine overlapping PE

Align reads to reference database

Clusterizereads to OTUs

Assign OTUs to taxonomy



Tools (examples):

Mothur
Qiime2
DADA2
MEGAN
MetaPhlAn4
PIPITS (for fungi)

Choosing the target of PCR amplification

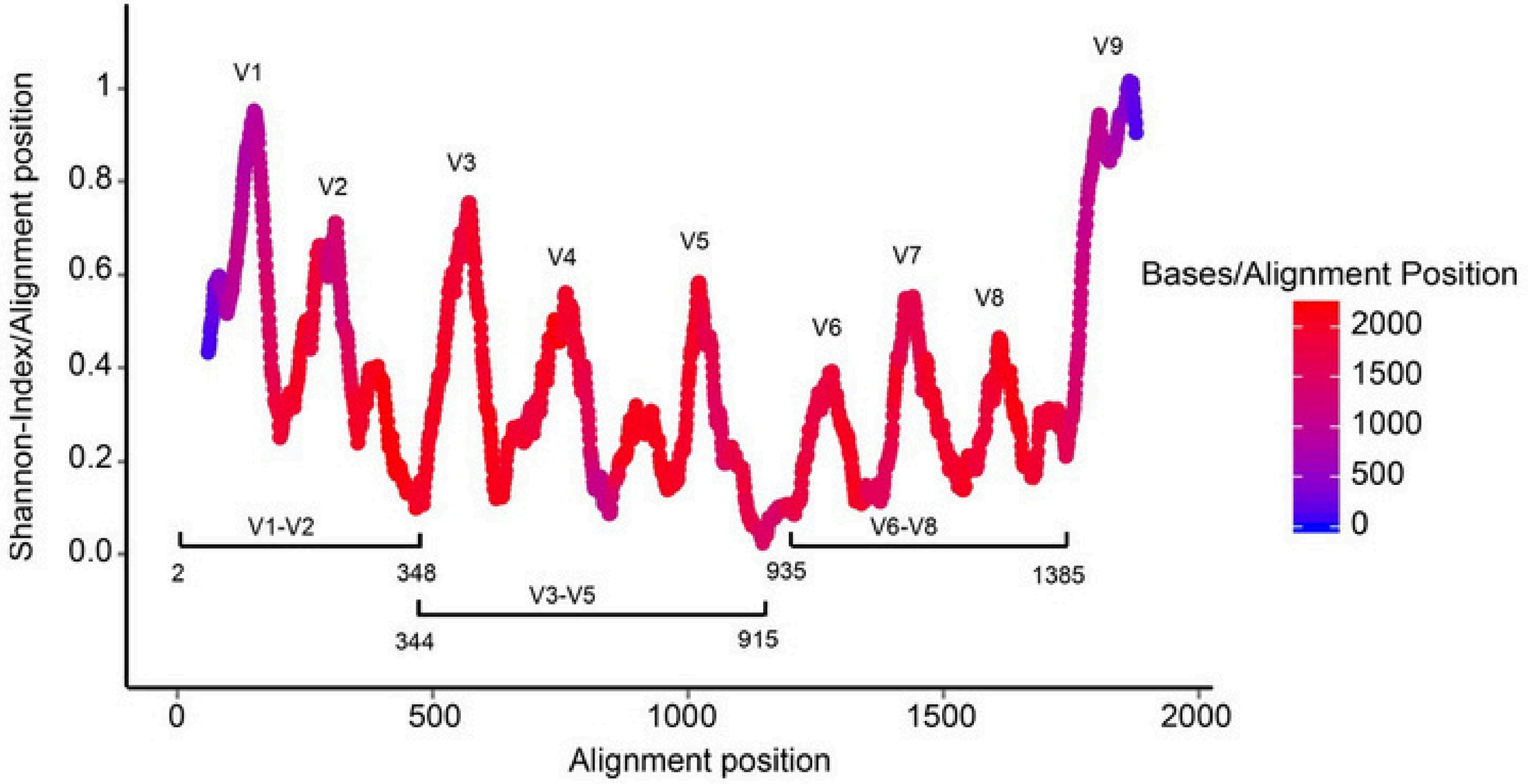
Usually this is a gene with some variable sequences in all species surrounded by conserved sequences:



Several genes have been used: 16S, 18S, ITS, gyrB, rpoB, recA, etc.

The most popular being the 16S for bacteria.

Selection of primers: reminder about 16S rRNA variable regions



Selection of the primers example 16S rRNA

primers:

799F

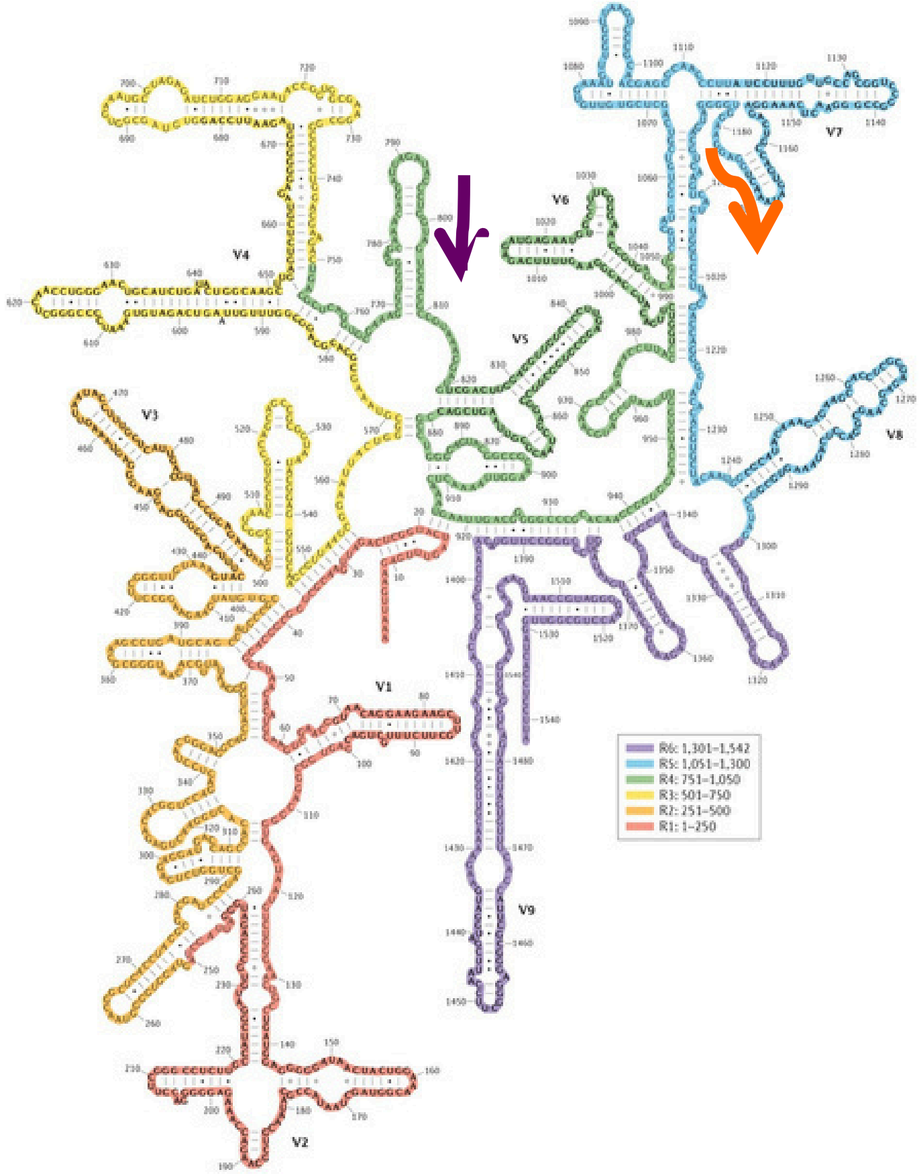
1193R

distance ~400bp

Sequenced by MiSeq 2x

300bp -> overlap!

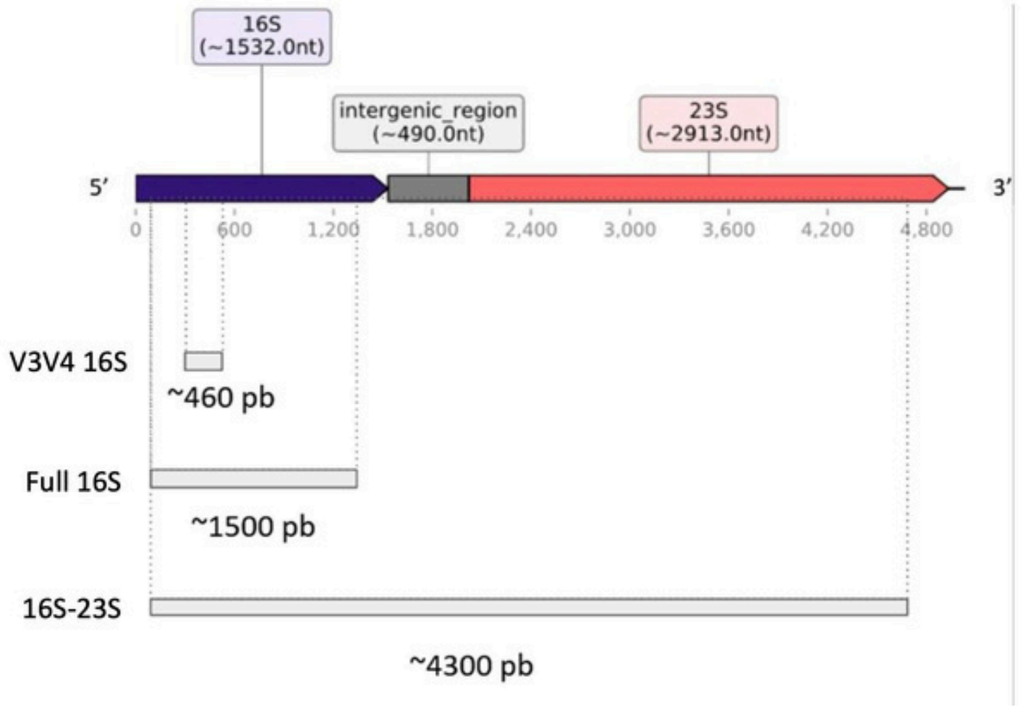




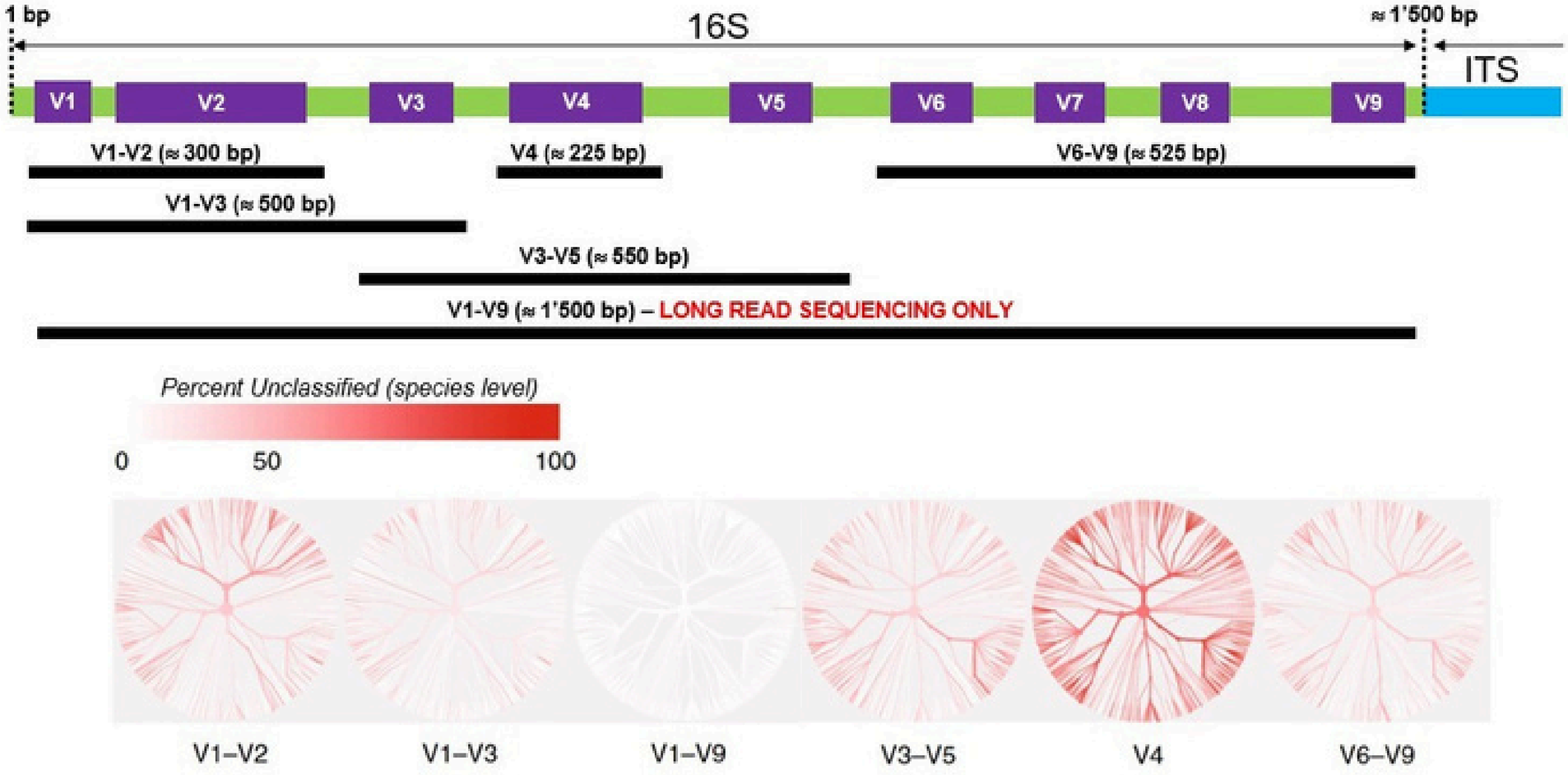
Nature Reviews | Microbiology

Full length amplicon sequencing

With the improvement of long read sequencing and the development of other long read techniques like LoopSeq, it becomes possible to sequence full length 16S or even larger amplicons.



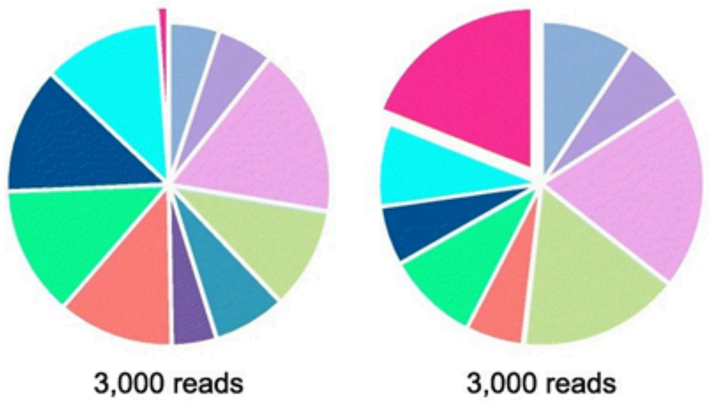
Improved classification with full length16S



Best with full length!

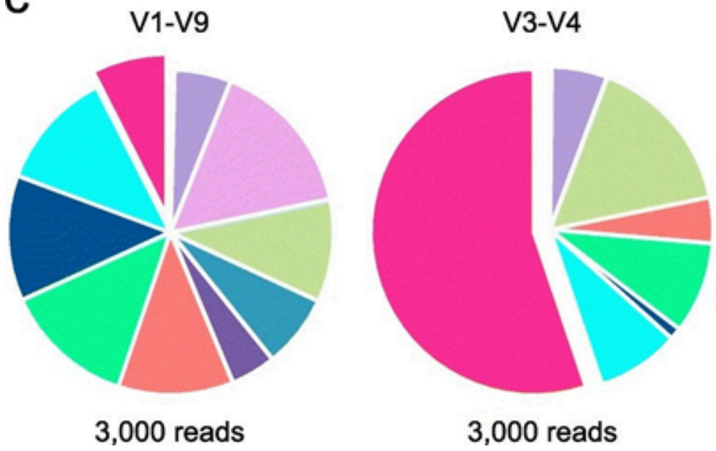
Example MOCK analysis withfull length16S vs V3-V4 (10 strains even mix)

b



Genus	V1-V9	V3-V4
Bacillus	5.07%	9.30%
Bifidobacterium	5.63%	6.60%
Clostridium	16.97%	19.70%
Deinococcus	10.23%	16.10%
Enterococcus	7.30%	0.13%
Escherichia	4.50%	0.00%
Lactobacillus	11.63%	5.77%
Rhodobacter	12.97%	9.17%
Staphylococcus	12.73%	5.97%
Streptococcus	11.87%	8.30%
mis- or unclassified	1.10%	18.97%

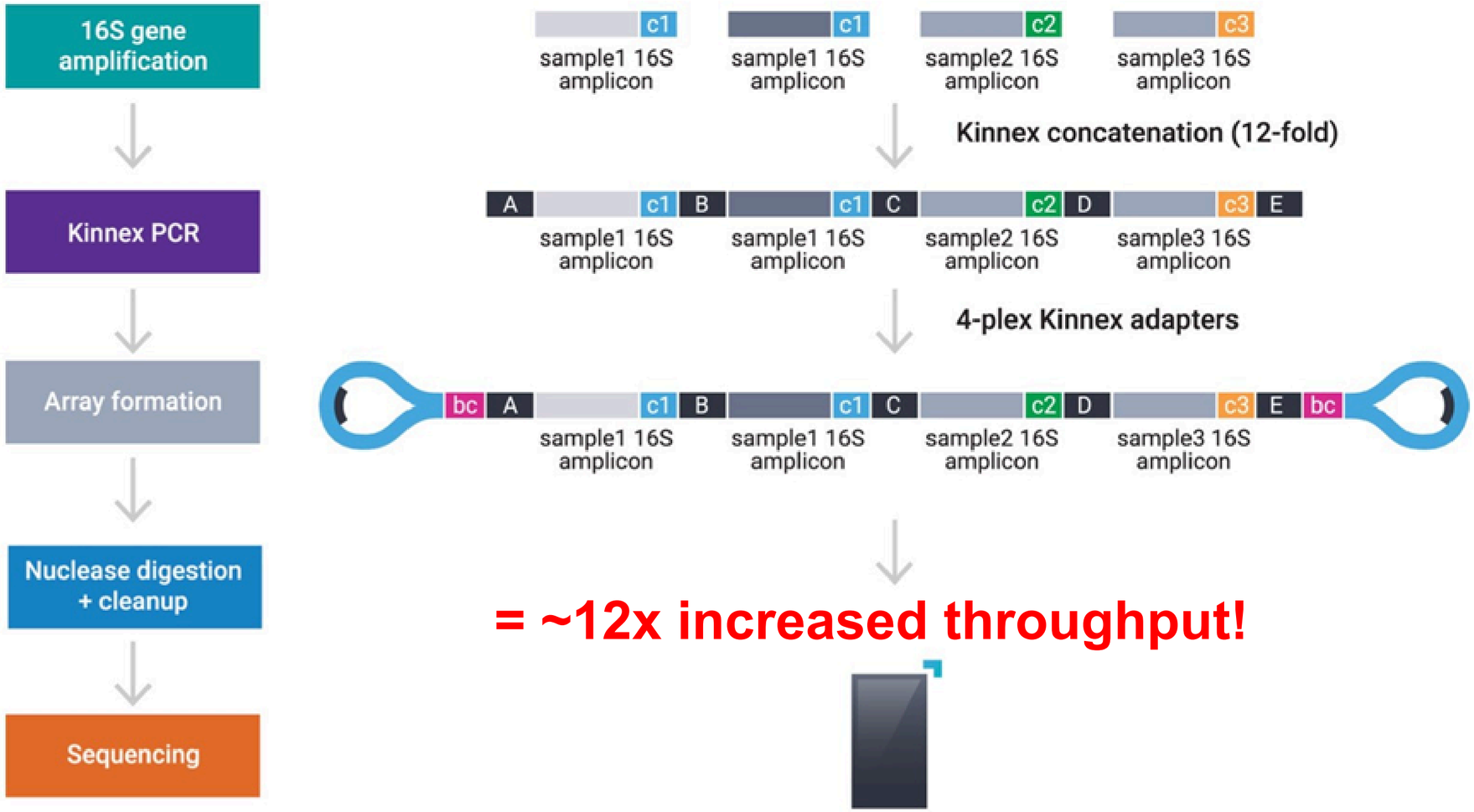
c



Species	V1-V9	V3-V4
B. cereus	0.40%	0.17%
B. adolescentis	5.63%	5.53%
C. beijerinckii	15.67%	0.03%
D. radiodurans	10.23%	16.07%
E. faecalis	7.30%	0.00%
Escherichia spp.	4.50%	0.00%
L. gasseri	11.47%	4.57%
R. sphaeroides	12.97%	9.07%
S. epidermidis	12.50%	1.20%
S. mutans	11.87%	8.20%
mis- or unclassified	7.47%	55.17%

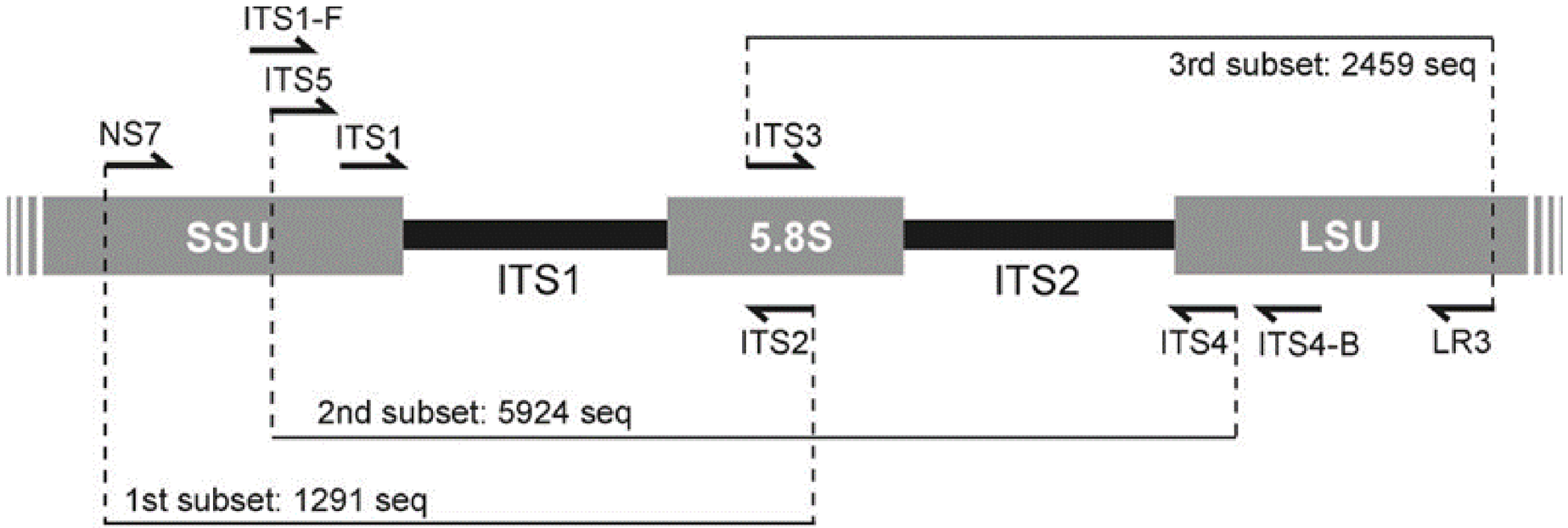
With PacBio: Kinnex16S rRNA library

Combining multiple full length 16S (up to 12) in one HiFi read!

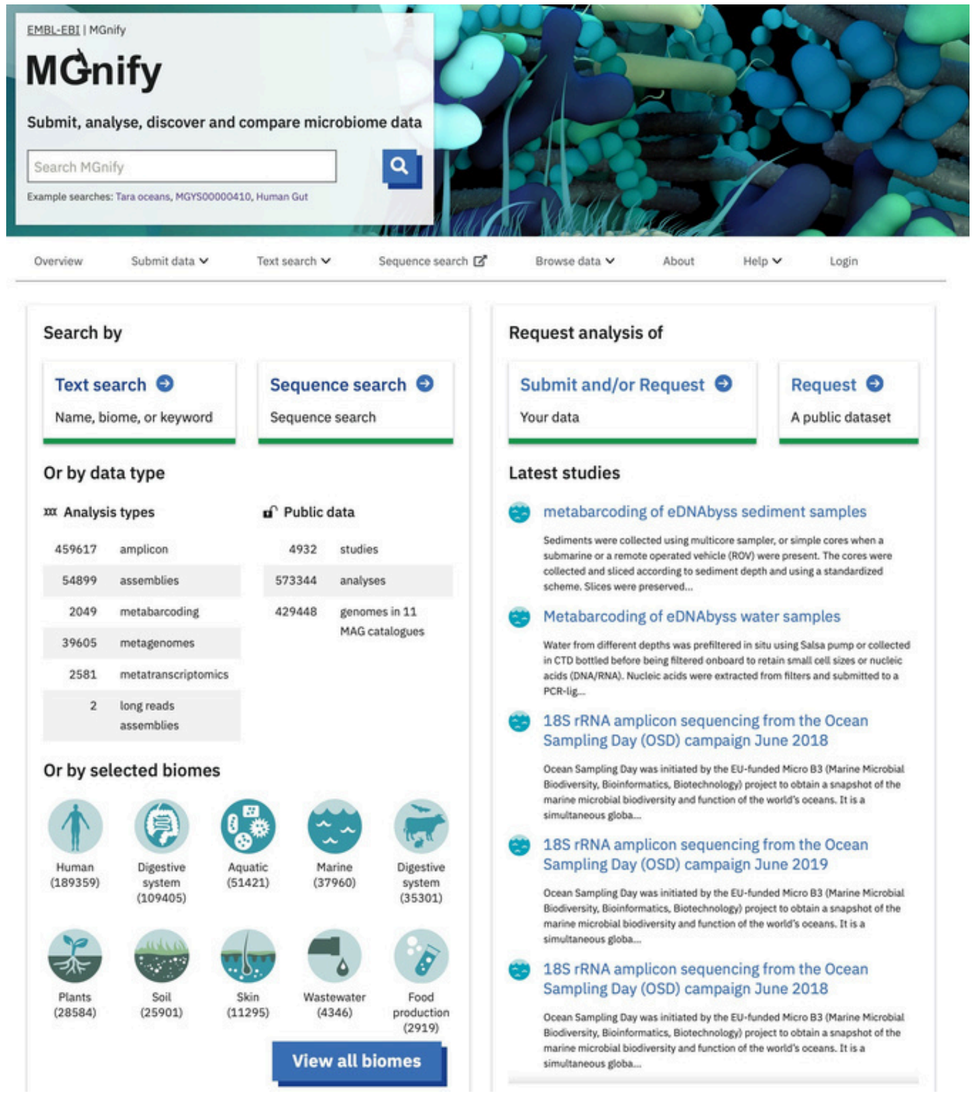


Internal Transcribed Spacers (ITS)

For fungi, the most widely used fungal genetic markers are the Internal Transcribed Spacers (ITS1 and ITS2) of the nuclear ribosomal subunit, which are located between the small and large subunit gene (SSU/18S and LSU/28S, respectively).

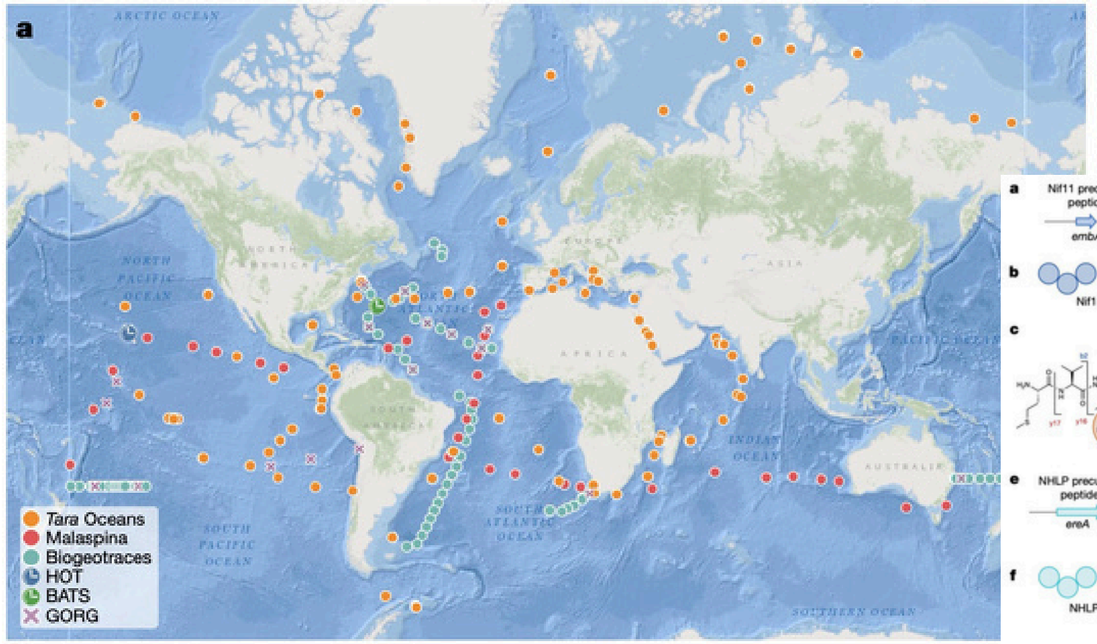


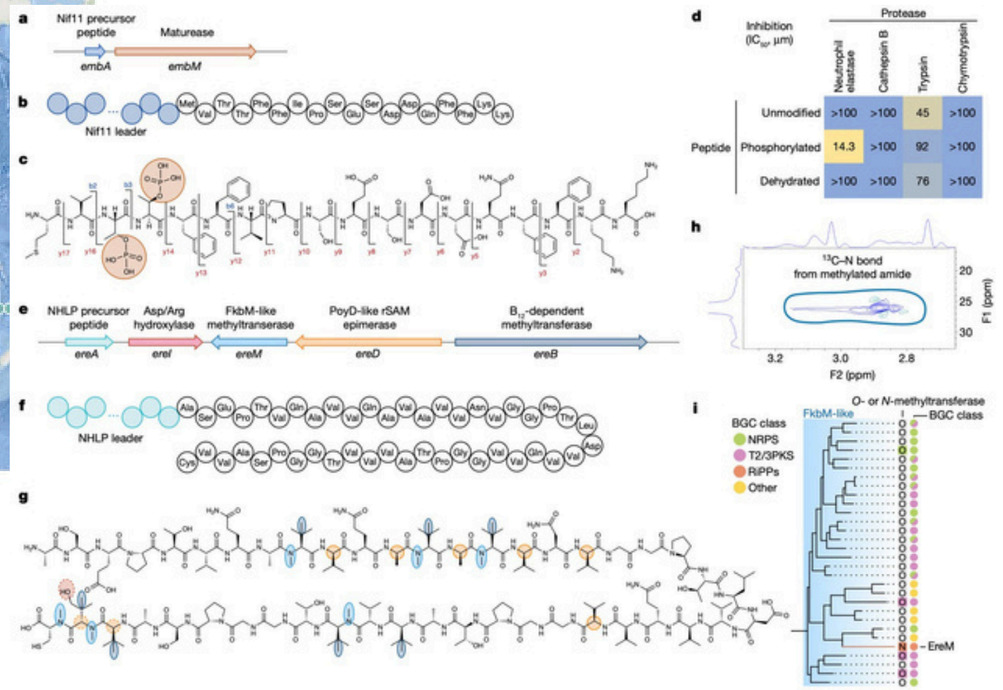
MGnify: the database available from EBI

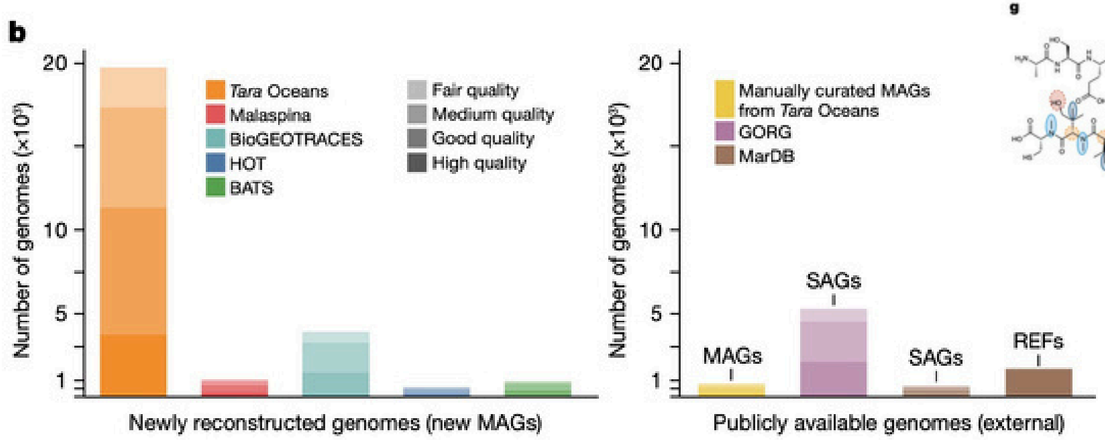


https://www.ebi.ac.uk/metagenomics

Ocean metagenome & pathways







https://microbiomics.io/ocean/

Summary

Method	Targeted	WGS
Short reads	**Currently standard** Many biases	**Currently standard** Several pipelines to choose from
Long reads (hifi)	Becomes competitive Many advantages Added throughput by Kinnex	Still expensive But much easier to obtain MAGs
Hybrid (short+long)	Not used in practice	Rarely combined

Summary

Importance of metagenomes for understanding our environment

Two main methods exist: shotgun vs targeted

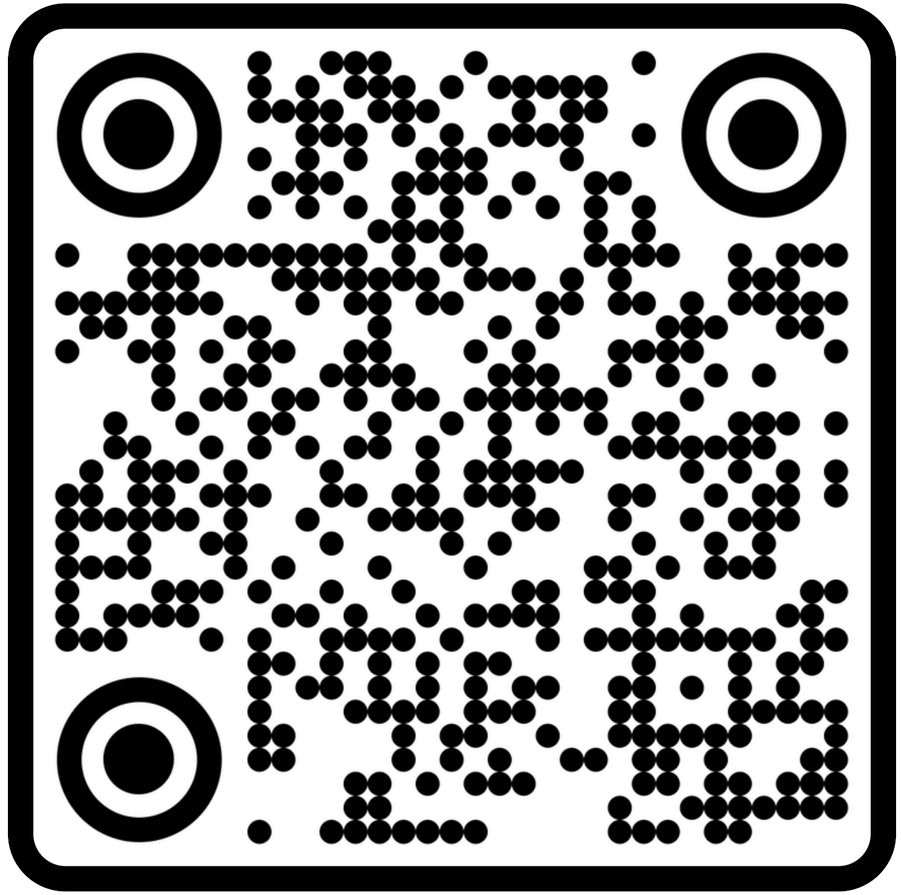
Both methods extract different things from the data

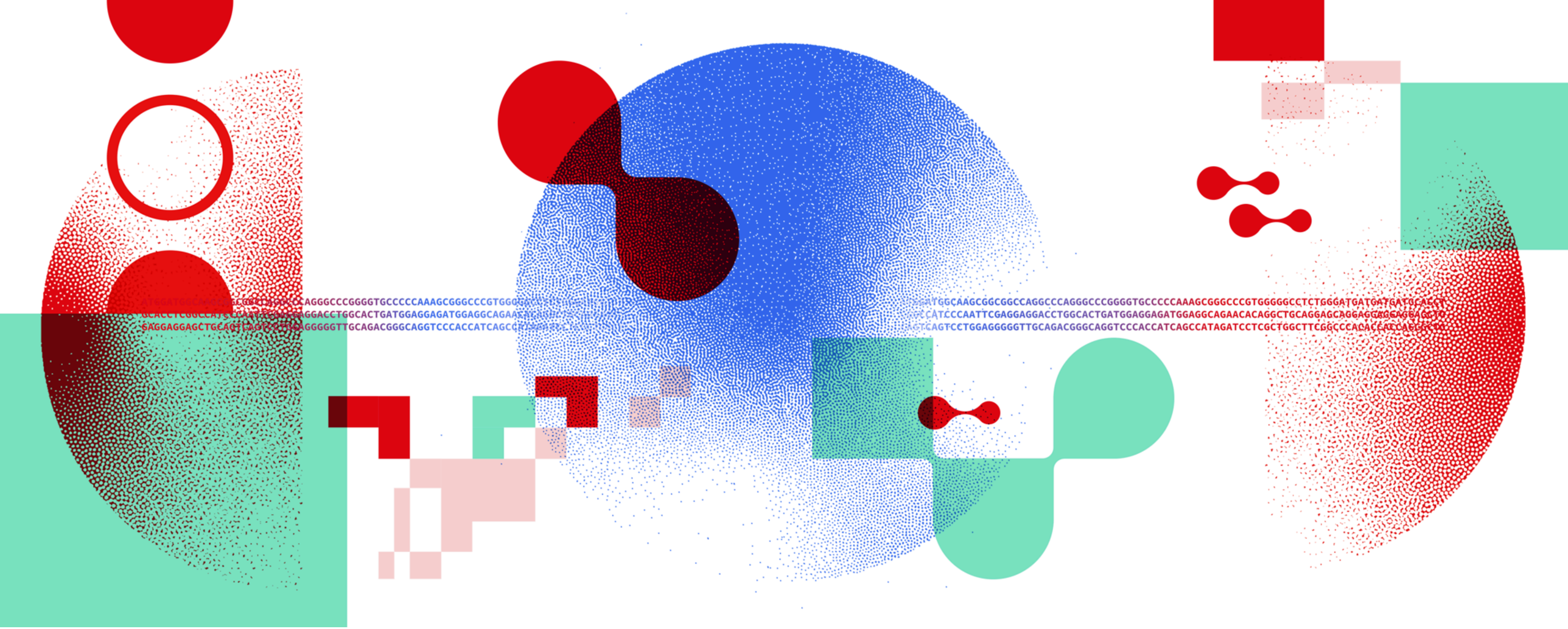
e.g., taxonomy assignment or functional assignment

Two sequencing techniques compete: short reads vs long reads

Many initiatives and databases gathering data

Practicals





Thank you

elixir SWITZERLAND

SIB